

EQUIP-FAM — Equipment Familiarisation

Platform and Component Familiarisation across Land, Naval, Air and Unmanned Systems
Training publication · Issue 1 · For instructional use

Purpose

This publication supports component familiarisation on six representative platforms. Each lesson covers exactly one platform and is intended to be delivered alongside the interactive 3D model of that platform, so that a trainee can select any labelled component on the model and read a grounded explanation of what it is and what it does.

Training objectives

- Identify by name every major component of each of the six platforms covered.
- State the function of each component and relate it to the platform's operational role.
- Distinguish land, naval, air and unmanned platform families by their characteristic assemblies.
- Use the interactive 3D model of a platform to locate a named component and describe it.

Course structure — one platform and one 3D model per lesson

The material is organised as three modules of two lessons each. Every lesson maps to exactly one preloaded 3D model and no model is used by more than one lesson.

- **Lesson 1.1 — Small-arms trainer (AK-pattern rifle)** (Land Systems). 3D model: Small-arms trainer (AK-pattern rifle). Focus: Weapon handling — parts, assembly, safety.
- **Lesson 1.2 — Main battle tank (Ariete C1 pattern)** (Land Systems). 3D model: Main battle tank (Ariete C1 pattern). Focus: Armoured corps — hull, turret, running gear.
- **Lesson 2.1 — Fast attack craft** (Naval Systems). 3D model: Fast attack craft. Focus: Naval familiarisation — armament, bridge, torpedo tubes.
- **Lesson 2.2 — Aircraft carrier (Kuznetsov pattern)** (Naval Systems). 3D model: Aircraft carrier (Kuznetsov pattern). Focus: Ship familiarisation — flight deck, island, armament.
- **Lesson 3.1 — Medium-altitude UAV (Rustom-2 pattern)** (Air And Unmanned Systems). 3D model: Medium-altitude UAV (Rustom-2 pattern). Focus: Unmanned systems — airframe, payload, propulsion.
- **Lesson 3.2 — Carrier-borne fighter (F-14 pattern)** (Air And Unmanned Systems). 3D model: Carrier-borne fighter (F-14 pattern). Focus: Airframe familiarisation — major assemblies.

How to use this document

Upload this document as course material, then attach the named 3D model to its lesson in the course review screen. Each lesson page carries a lesson body that summarises the platform and names every component taught on it, followed by a component reference giving the detail behind each name.

Lesson 1.1 — Small-arms trainer (AK-pattern rifle)

3D model for this lesson: **Small-arms trainer (AK-pattern rifle)** (model key: rifle-trainer). Suitable for: Weapon handling — parts, assembly, safety. This model is used by this lesson only.

Lesson body — reproduce this text in full

The small-arms trainer is an AK-pattern, gas-operated rifle used for weapon-handling, assembly and safety drills, and five components are taught on it. The barrel is the rifled tube that directs the projectile and contains the propellant gases at the moment of firing, while the gas tube / handguard bleeds gas from the barrel port to drive the action and gives the firer an insulated grip. The magazine is the detachable box that holds cartridges and feeds them upward under spring pressure into the path of the bolt. The bolt carrier is the reciprocating group that strips a round from the magazine, locks and unlocks the bolt and extracts and ejects the spent case, and the stock is the shoulder-braced furniture at the rear that stabilises the weapon and absorbs recoil. In handling drills the rifle is proved clear by removing the magazine first and then retracting the bolt carrier to inspect the chamber through the ejection port. First-line checks cover bore condition at the barrel, security of the gas tube / handguard, the lips and spring of the magazine, free travel of the bolt carrier, and tightness of the stock at its attachment. Stripping for cleaning follows the same order, so a trainee who can name these five components in sequence can also strip, inspect and reassemble the weapon.

Component reference

Barrel — The rifled tube through which the projectile is driven. It contains the propellant gases, imparts spin for stability, and carries the gas port that taps a portion of those gases to operate the action. Barrel condition and bore cleanliness are first-line inspection items.

Gas tube / handguard — The gas tube carries propellant gas tapped at the barrel port rearward onto the piston, which drives the bolt carrier to the rear and cycles the weapon. The handguard encloses it, insulating the firer's supporting hand from a hot barrel and providing the forward grip.

Magazine — The detachable curved box that stores cartridges and presents them one at a time to the bolt under spring pressure from the follower. It latches into the magazine well and is the item most often responsible for feed stoppages, so it is checked for lip damage and spring fatigue.

Bolt carrier — The reciprocating assembly that carries the bolt. On its forward stroke it strips a round from the magazine, chambers it and rotates the bolt into lock; on its rearward stroke it unlocks, extracts and ejects the spent case and re-cocks the hammer. It is the heart of the cycle of operation.

Stock — The shoulder-braced furniture at the rear of the weapon. It sets the length of pull, gives the firer a stable third point of contact, absorbs and directs recoil into the shoulder, and on this pattern encloses the return spring assembly.

Lesson 1.2 — Main battle tank (Ariete C1 pattern)

3D model for this lesson: **Main battle tank (Ariete C1 pattern)** (model key: main-battle-tank). Suitable for: Armoured corps — hull, turret, running gear. This model is used by this lesson only.

Lesson body — reproduce this text in full

The Ariete C1-pattern main battle tank is a four-crew armoured fighting vehicle taught in six major components. The main gun is the 120 mm smoothbore primary armament that fires the tank's kinetic-energy and high-explosive natures, and the turret is the rotating armoured structure that carries it and traverses through 360 degrees to lay onto a target. The commander's cupola is the raised all-round-vision station on the turret roof from which the commander observes, designates targets and fights the vehicle. The track & running gear — road wheels, sprocket, idler and torsion-bar suspension — carries the hull's weight and delivers mobility over broken ground, the glacis / hull front is the steeply sloped frontal armour that presents the greatest effective thickness to incoming fire, and the engine deck at the rear covers the powerpack, housing the engine, transmission and cooling group. Crew duties map directly onto these components. The gunner lays the main gun using the turret's powered traverse and elevation; the commander fights the vehicle from the commander's cupola with all-round vision; the driver sits in the hull behind the glacis / hull front, the vehicle's thickest protection. First-parade checks concentrate on track tension and wheel condition in the track & running gear and on keeping the engine deck grilles clear so the powerpack can breathe.

Component reference

Main gun — The 120 mm smoothbore primary armament, mounted in the turret on trunnions and stabilised in two axes so it can be fired accurately on the move. It fires kinetic-energy and high-explosive natures against armour and soft targets, and its bore evacuator and thermal sleeve are routine inspection points.

Turret — The rotating armoured structure above the hull that carries the main gun, the gunner's and commander's sights and the ready ammunition. Powered traverse through 360 degrees lets the crew engage in any direction independently of the hull's heading, with a manual back-up for use if power is lost.

Commander's cupola — The raised station in the turret roof over the commander's position, fitted with all-round vision blocks and an independent panoramic sight. From it the commander maintains local awareness, designates targets to the gunner and can override the turret to lay the gun himself.

Track & running gear — The continuous tracks together with the road wheels, drive sprocket, idler, return rollers and torsion-bar suspension. This group spreads the vehicle's weight to give low ground pressure, transmits drive to the ground and absorbs terrain shock. Track tension and pin wear are daily first-line checks.

Glacis / hull front — The steeply sloped frontal armour plate of the hull. The slope increases the line-of-sight thickness a projectile must pass through and encourages ricochet, making it the vehicle's best protected aspect — which is why crews are taught to keep the glacis toward the threat.

Engine deck — The armoured decking over the rear hull that covers the powerpack: the engine, transmission, cooling group and air filtration. It is built as removable panels and grilles so the complete powerpack can be lifted out for replacement, and its grilles must be kept clear for cooling airflow.

Lesson 2.1 — Fast attack craft

3D model for this lesson: **Fast attack craft** (model key: patrol-boat). Suitable for: Naval familiarisation — armament, bridge, torpedo tubes. This model is used by this lesson only.

Lesson body — reproduce this text in full

The fast attack craft is a small, high-speed surface combatant that trades protection for speed and strike power, and six components are taught on it. The main gun (forward) is the primary bow-mounted gun that engages surface and air targets across the forward arc, and the main gun (aft) is the matching stern mount that covers the rear arc and protects the craft while it withdraws. The auxiliary gun is the lighter, faster-traversing secondary weapon used against close-range small craft and low-flying air threats. The torpedo tubes are the deck-mounted launchers that fire the craft's anti-ship torpedoes on a firing solution passed from the operations room, the mast / radar carries the surface-search and navigation antennas that generate that picture, and the bridge / superstructure is the enclosed control position from which the craft is conned, navigated and fought. In a typical engagement these components work as one system. The mast / radar detects and tracks the contact, the bridge / superstructure assesses the picture and directs the attack, the torpedo tubes and the main gun (forward) deliver the strike as the craft closes, and the main gun (aft) and the auxiliary gun cover the high-speed withdrawal. The craft's survival depends on speed and first detection rather than on armour.

Component reference

Main gun (forward) — The primary gun mounting on the forecastle, forward of the superstructure. It has the widest unobstructed arc of the craft's weapons and is the main battery for engaging surface contacts and air targets ahead and on either bow during a closing run.

Main gun (aft) — The matching gun mounting on the quarterdeck. It covers the arcs the forward gun cannot reach and is the weapon that keeps a pursuer under fire while the craft uses its speed to break contact after a strike.

Auxiliary gun — A lighter, higher-rate-of-fire secondary mounting with faster traverse than the main guns. It is used for close-range self defence against fast inshore attack craft, small boats and low, slow air targets, and for warning fire during boarding and interdiction tasks.

Torpedo tubes — Deck-mounted launch tubes carrying the craft's anti-ship torpedoes. They are trained onto the target bearing and fired on a solution computed in the operations room from radar and passive data. The torpedo is the craft's heaviest punch, letting a small hull threaten a major unit.

Mast / radar — The lattice mast amidships carrying the surface-search, navigation and fire-control antennas together with the electronic-support and identification aerials. Its height sets the radar horizon, so it is the sensor that builds the tactical picture the guns and torpedoes are fired on.

Bridge / superstructure — The enclosed deckhouse containing the bridge, the operations room and the craft's control positions. From here the craft is conned and navigated, the sensor picture is assessed and the weapons are directed; it also carries the mast and the main communications fit.

Lesson 2.2 — Aircraft carrier (Kuznetsov pattern)

3D model for this lesson: **Aircraft carrier (Kuznetsov pattern)** (model key: aircraft-carrier). Suitable for: Ship familiarisation — flight deck, island, armament. This model is used by this lesson only.

Lesson body — reproduce this text in full

The Kuznetsov-pattern aircraft carrier is an aviation-capable capital ship whose flight deck, island and armament make up six taught components. The ski-jump ramp is the upward-curved forward section of the flight deck that gives a launching aircraft the pitch and vertical velocity to get airborne without a catapult, and the P-700 launcher deck forward of the flight deck holds the ship's long-range anti-ship missile cells beneath flush hatches. The island / superstructure is the starboard-side tower carrying the bridge, flying control and the main sensor masts, from which both the ship and its air operations are controlled, while the hangar deck beneath the flight deck is the enclosed space where aircraft are stowed, armed and maintained before the lifts move them topside. The Kashtan CIWS is the close-in weapon system that combines rapid-fire guns and short-range missiles on a single mount to destroy leakers penetrating the outer air defence, and the RBU-12000 launcher is the multi-barrelled rocket launcher used for anti-submarine and anti-torpedo defence. In flight operations the components work in sequence. Aircraft are prepared on the hangar deck, raised by the deck lifts to the flight deck, and launched over the ski-jump ramp under direction from flying control in the island / superstructure. Independently of the air group, the P-700 launcher deck provides long-range anti-ship fire, while the Kashtan CIWS and the RBU-12000 launcher defend the ship against leaking air threats and against submarines and torpedoes.

Component reference

Ski-jump ramp — The upward-curved forward end of the flight deck. An aircraft leaving the ramp is thrown into a ballistic arc with positive pitch and vertical velocity, buying the seconds it needs to accelerate to flying speed. It is what allows fixed-wing launch from a deck with no catapult.

P-700 launcher deck — The section of deck forward of the take-off run under which the ship's long-range anti-ship missile cells are recessed, closed by flush hatches so aircraft can taxi over them. It gives the carrier an organic surface-strike capability independent of her air wing.

Island / superstructure — The tower offset to starboard so it does not obstruct the flight deck. It houses the navigating bridge, flying control and the air operations positions, and carries the main radar arrays, funnel uptakes and communications aerials, making it the ship's command centre.

Hangar deck — The large enclosed deck below the flight deck where aircraft are stowed, armed, refuelled and maintained out of the weather. Aircraft move between it and the flight deck on deck-edge lifts, and its size and layout set the practical limit on the size of the embarked air group.

Kashtan CIWS — The close-in weapon system: a single stabilised mount combining rapid-fire rotary cannon with short-range surface-to-air missiles and its own radar and electro-optical director. It is the last-ditch layer of air defence, engaging sea-skimming missiles and aircraft that have penetrated the outer screen.

RBU-12000 launcher — A multi-barrelled rocket launcher mounted forward. It fires salvoes of rockets to lay a barrage in the water against submarine contacts and, in its anti-torpedo role, to decoy or destroy incoming torpedoes before they reach the hull.

Lesson 3.1 — Medium-altitude UAV (Rustom-2 pattern)

3D model for this lesson: **Medium-altitude UAV (Rustom-2 pattern)** (model key: uav-rustom). Suitable for: Unmanned systems — airframe, payload, propulsion. This model is used by this lesson only.

Lesson body — reproduce this text in full

The Rustom-2-pattern medium-altitude long-endurance UAV is an unmanned surveillance aircraft flown from a ground control station, and five components are taught on it. The nose / sensor payload is the gimballed electro-optical and infrared turret beneath the forward fuselage that provides the day-and-night imagery the mission is flown for. The fuselage / avionics bay is the central airframe structure carrying the flight-control computers, datalink and navigation equipment while transferring loads between the wing and the tail. The engine nacelle houses the propulsion engine and its accessories and mounts it clear of the airframe, and the wing is the high-aspect-ratio lifting surface whose long span gives the efficiency needed for endurance at altitude and whose control surfaces provide roll control. The tail / stabiliser at the rear provides pitch and yaw stability and carries the surfaces that trim the aircraft in flight. On a sortie the aircraft is commanded from a ground control station through the datalink housed in the fuselage / avionics bay. The engine nacelle and the wing between them sustain the aircraft at medium altitude for many hours, the tail / stabiliser holds it steady on track while the autopilot flies a search pattern, and the nose / sensor payload delivers the imagery back to the station in real time. Endurance, not speed, is the design driver throughout.

Component reference

Nose / sensor payload — The gimballed turret under the nose carrying the electro-optical and infrared cameras, and often a laser designator or range finder. Stabilised against airframe motion, it is the reason the sortie is flown: it delivers the day-and-night imagery streamed back over the datalink.

Fuselage / avionics bay — The central body of the airframe. It carries the flight-control computers, the navigation set, the satellite and line-of-sight datalink equipment and the fuel, and structurally ties the wing to the tail. Access panels here are the main points for pre-flight avionics checks.

Engine nacelle — The streamlined housing that carries the propulsion engine and its accessories, mounted clear of the fuselage to keep exhaust and vibration away from the airframe and sensors. It contains the engine mounts, cooling intakes and the fire-detection provision.

Wing — The high-aspect-ratio lifting surface, and the longest dimension of this airframe. The long, slender span produces lift with very little induced drag, which is what gives a MALE UAV its endurance; its ailerons and flaps provide roll control and low-speed lift for take-off and landing.

Tail / stabiliser — The empennage at the rear of the fuselage, on this pattern arranged as a V-tail. It provides the pitch and yaw stability that keeps the aircraft tracking straight, and its moving surfaces trim the aircraft and give the autopilot its control authority.

Lesson 3.2 — Carrier-borne fighter (F-14 pattern)

3D model for this lesson: **Carrier-borne fighter (F-14 pattern)** (model key: fighter-aircraft). Suitable for: Airframe familiarisation — major assemblies. This model is used by this lesson only.

Lesson body — reproduce this text in full

The F-14-pattern carrier-borne fighter is a two-seat, twin-engined interceptor designed to operate from a carrier deck, and five components are taught on it. The nose / radome is the dielectric forward cone that covers and protects the fire-control radar antenna while letting its beam pass, and the cockpit immediately behind it is the two-seat crew station housing the pilot forward and the radar intercept officer aft with their displays, controls and ejection seats. The variable-geometry wing is the pivoting swing wing that sweeps forward for lift at the low approach speeds needed to recover to a deck and sweeps aft to cut drag at high speed. The vertical stabiliser, fitted here as a twin-fin arrangement, provides directional stability and carries the rudders for yaw control, and the engine / exhaust group at the rear produces the thrust for catapult launch, intercept and afterburning acceleration. On recovery to the deck these components work together. The variable-geometry wing sweeps forward to give the lift needed at approach speed, the crew in the cockpit fly the pass on instruments with the fire-control radar behind the nose / radome supporting the intercept beforehand, the vertical stabiliser holds the aircraft straight in the turbulent air astern of the ship, and the engine / exhaust group is held at high power throughout in case the hook misses and a bolter is flown.

Component reference

Nose / radome — The dielectric cone at the front of the aircraft. It is shaped for low drag and built of a material transparent to radio frequency energy so the fire-control radar behind it can transmit and receive through it while the antenna is protected from airflow, rain and debris.

Cockpit — The two-seat crew station under a single canopy: the pilot forward flying the aircraft, and the radar intercept officer aft working the radar and weapons. It contains the flight and engine instruments, the displays, the control column and throttles, and the ejection seats for both crew.

Variable-geometry wing — The pivoting swing wing. Swept forward it gives a long span, high lift and docile handling for carrier approach and landing; swept aft it reduces span and drag for supersonic dash. The sweep is scheduled automatically with speed, and it is what lets one airframe do both jobs.

Vertical stabiliser — The fin — twin fins on this pattern, canted outboard — that provides directional (yaw) stability and keeps the aircraft weathercocked into the relative airflow. It carries the rudder, and the twin arrangement keeps each fin short enough for hangar-deck clearance.

Engine / exhaust — The propulsion group in the rear fuselage: the engines, their intakes and the exhaust nozzles. It delivers the thrust for catapult launch, climb and intercept, and the afterburners inject fuel into the exhaust for the extra thrust needed at launch and in combat.

Glossary

Component familiarisation — Instruction in which a trainee identifies each major assembly of a platform by name, states its function and relates it to the platform's role. It is the first step before any maintenance, handling or operating instruction.

Hotspot — A labelled point placed on a 3D model that corresponds to one taught component. Selecting a hotspot in the viewer calls up the grounded explanation of that component from the course material.

Cycle of operation — The repeating sequence a self-loading weapon performs for each shot: feed, chamber, lock, fire, unlock, extract, eject and cock. On a gas-operated rifle it is driven by gas tapped from the barrel acting on the bolt carrier.

Gas operation — A method of self-loading in which a portion of the propellant gas is tapped from the barrel through a gas port and used to drive the action to the rear. It is the operating principle of the AK-pattern small-arms trainer.

Glacis — The steeply sloped frontal armour plate of an armoured vehicle's hull. The slope increases the effective thickness a projectile must penetrate and promotes ricochet, making it the best protected aspect of the vehicle.

Close-in weapon system (CIWS) — A short-range, self-contained air-defence mount combining rapid-fire guns, and sometimes missiles, with its own radar and director. It is the last layer of defence against anti-ship missiles and aircraft that penetrate the outer screen.

Ski-jump launch — Launching fixed-wing aircraft from a curved ramp at the bow instead of a catapult. The ramp imparts pitch and vertical velocity so the aircraft can accelerate to flying speed after leaving the deck.

Variable geometry — A wing whose sweep angle can be changed in flight. Low sweep gives lift and docile handling at approach speeds; high sweep reduces drag for high-speed flight, letting one airframe cover both regimes.